Lab 7: Polished report, weak citations: route information, not just tasks

Topic 1: Agentic Architecture & Orchestration · Task 1.3: Configure subagent invocation, context passing, and spawning

Runs on Simulator 3: Multi-Agent Research System · Self-practice

Source: SkillCertPro practice exam Part 1, Q10 (unofficial third-party question)

Commit to an answer first

After several runs, the final reports contain polished summaries but weak citation coverage. The web and document subagents return useful source details to the coordinator, but the synthesis subagent is invoked later with only a brief instruction to produce themes and recommendations. What change would best improve the orchestration design?

A. Include the relevant findings and source metadata directly in the synthesis subagent's Agent prompt.

B. Allow research subagents to communicate directly so they can share source details before synthesis.

C. Broaden the synthesis subagent's description so automatic routing selects it after research completes.

D. Convert the workflow to independent agent-team sessions with a shared task list for every report.

Simulate the solution

```
# In the browser: open Simulator 3, then Lab 7
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 7

# In a terminal, from simulators/simulator-3-multi-agent-research/
uv run research_system.py --case UC-10
```

What you should see

The synthesis prompt runs to thousands of characters, with URLs, document names and dates.

The Agent prompt sent to synthesis is large (thousands of characters) and contains URLs, document names and dates from the research phase.

Open `simulators/simulator-3-multi-agent-research/evidence/` and find it yourself.

The answer

A. Put findings and source metadata directly in the synthesis prompt.

In a coordinator–worker design the coordinator routes information, not just tasks. Subagents keep their own context windows, which stops intermediate output from flooding the parent, but that same isolation means the synthesis subagent never saw what the search and document subagents saw. The parent-to-subagent channel is the Agent tool prompt string, so excerpts, URLs, document names, publication dates and claim-to-source mappings belong there. This preserves provenance without changing the hub-and-spoke pattern.

Why the others fail

B. Documented subagents do not talk to each other. They return results to the main conversation.

C. A subagent's description affects when Claude delegates, not what context is passed.

D. Agent teams (independent sessions with inter-agent messaging) are a heavier, different pattern. That is a redesign to fix a missing-context problem.

Check yourself

- synthesis packet >= 2000 characters
- synthesis packet contains at least 3 source URLs or document names
- synthesis packet contains publication dates

Now change one thing

In fixtures/sources.json replace the visual-arts TIMEOUT with [] and rerun: a valid empty result must not read as a failure.

Third-party practice material from SkillCertPro, reconstructed from a video transcript in the Tertiary Markdown Vault. Not a real exam item. Wording may differ from the original.